

Amin Ahmed G

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PROFESSIONAL SUMMARY

Detail-oriented Robotics & Automation engineering student (Expected 2027) with practical experience in ROS 2, Gazebo simulations, and low-latency systems programming in Rust and C++. Proven capability in designing real-time network watchdogs, tuning PID control loops on STM32 microcontrollers, and deploying warehouse automation solutions.

EDUCATION

B.E. Robotics and Automation — Anna University

Expected 2027

Coursework: Control Systems, Embedded C, Robot Kinematics, Sensors & Instrumentation, Microprocessors

TECHNICAL SKILLS

Frameworks: ROS 2 (Jazzy, Humble), Gazebo, Nav2, SLAM, MoveIt

Languages: C, C++, Python, Rust, Bash

Hardware: STM32, Raspberry Pi, ESP32, Lidar, PID Control, UART/I2C/SPI

Design: Fusion 360, SolidWorks, EasyEDA, KiCad

Tools: Linux, Docker, Git, STM32CubeIDE, CODESYS

PROJECTS

RoboShield: RTPS/DDS Security Watchdog — *Rust, libpcap*

2026 / Ongoing

- Developed a low-latency, out-of-band Network Intrusion Detection System (NIDS) in Rust to inspect raw RTPS/DDS packets.
- Designed a zero-allocation, endianness-aware parser that extracts participant GUIDs and topic ParameterLists from UDP streams.
- Achieved an average processing latency of **1.36 microseconds** ($\approx 734k$ packets/sec), supporting strict real-time control loops.
- Implemented sliding-window rate tracking to detect high-frequency topic floods and unauthorized publisher injection.

ROS 2 Mobile Robot Simulations — *mecanum_bot & two_wheel_robot*

2025

- Modeled, simulated, and configured controller architectures for differential drive and mecanum wheeled mobile robots in ROS 2 Jazzy and Gazebo Harmonic.
- Integrated LiDAR sensor plugins, configured joint state broadcasters and diff-drive controllers, and implemented Python-based waypoint followers.

Line Follower Robot — *STM32*

2024

- Built a line follower robot on STM32 with PID control and sensor fusion for consistent tracking at speed.
- Tuned chassis weight and traction; control loop runs under 1ms.

Temperature Sensor PCB — *ESP32 (EasyEDA)*

2025

- Designed a 2-layer PCB in EasyEDA with ESP32, AMS1117 regulator, LM35 sensor, USB power, and UART header — passed DRC clean.
- Exported Gerbers for JLCPCB — $\approx \$5.44$ for 5 boards.

Motor Control PLC — *CODESYS Ladder Logic*

2025

- Wrote ladder logic in CODESYS — start/stop with seal-in, normally-closed E-stop, and a 2-second TON timer.
- Built and ran clean (0 errors) on the CODESYS soft PLC runtime; tested all logic paths in simulation.

INTERNSHIPS

Robonetics Automation Solutions LLP — *Chennai*

2025 — 21 Days

- Worked with JAKA cobots, picked up SolidWorks basics, and saw how AGV/AMR systems are deployed in a real warehouse environment.

Drone Technology Internship — *NSIC Technical Services Centre, Chennai*

7 Days

- Assembled and disassembled an agricultural drone frame; did basic CATIA modeling and learned UAV construction fundamentals.

LEADERSHIP

Event Organizer — *Line Follower Robot Competition*

2024

- Wrote the rulebook and designed the track for our university LFR competition.

Linux System Optimization

Ongoing

- Set up and maintained my own Ubuntu/Hyprland environment tuned for robotics work — low latency, minimal overhead.